
Joint-CS: Simultaneous Illumination Normalization and Sparse Image Compression via Unfolded Recovery

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Abstract

The traditional camera pipeline samples at Shannon–Nyquist rates, compresses to a sparse representation, and only afterwards performs illumination correction on the decoded image. This sequential ordering discards information: sparse recovery applied to a signal whose dynamic range is dominated by an unknown multiplicative illumination field has weaker recovery guarantees than the same recovery applied to the underlying reflectance. We formulate *Joint-CS*, a single inverse problem in the joint variables of DCT-domain reflectance coefficients and a smooth per-column illumination field, and solve it by block-coordinate alternation between a learned-iterative-shrinkage step on the coefficients and a closed-form ridge step on the illumination. Across four experiments—an empirical Donoho–Tanner phase transition, rate-distortion on a natural image, a deep-unfolded LISTA solver, and an ablation against a sequential CS-then-illumination-correct baseline—we find that the joint formulation outperforms the sequential one by 1.4–2.9 dB PSNR across measurement rates under an $8\times$ horizontal illumination gradient, and that a $K=10$ unrolled LISTA matches what FISTA needs 22 iterations to reach.

1 Introduction

Modern imaging pipelines acquire N pixels via Nyquist sampling, then immediately throw most of that information away: a typical JPEG keeps only $\sim 5\text{--}10\%$ of the DCT coefficients above the quantization floor, and the image signal processor (ISP) further reshapes the data through demosaicing, tone-mapping, and white-balance correction. Compressed sensing (CS) reverses this order. If the signal $\mathbf{x}$ is S -sparse in some basis Ψ , then $M \ll N$ random linear measurements $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{w}$ suffice to recover $\mathbf{x}$ exactly (up to noise) by solving an ℓ_1 minimization, provided $\mathbf{A}$ obeys the restricted isometry property (3; 4).

CS has been extensively studied for compression. It has been less extensively studied as an integrated stage of a RAW image-signal pipeline. The crux of the issue is that a RAW sensor measures the product

$$\mathbf{x} = \mathbf{s} \odot \mathbf{g}, \tag{1}$$

where $\mathbf{s}$ is the underlying scene reflectance (sparse in the DCT) and $\mathbf{g}$ is a spatially-varying illumination field (smooth, not sparse). A sequential pipeline—CS recovery of $\mathbf{x}$, then a downstream divide-by-illumination step—wastes the prior that $\mathbf{s}$, *not* $\mathbf{x}$, is sparse in Ψ . Multiplying a DCT-sparse signal by an arbitrary illumination field destroys the sparsity that the ℓ_1 regularizer is trying to exploit; the recovered $\mathbf{x}$ then carries an illumination-shaped bias that no downstream column-mean correction can fully undo.

Contributions. We make three contributions:

1. We formulate *Joint-CS*: a joint ℓ_1 /smooth-prior objective in $(\mathbf{c}, \mathbf{g})$ that performs sparse recovery and illumination correction in a single optimization, and we derive a block-coordinate solver with a FISTA inner loop on $\mathbf{c}$ and a closed-form ridge step on $\mathbf{g}$.
2. We deep-unfold the $\mathbf{c}$ -step into a learned $K=10$ -layer LISTA network (6) and stabilize its training under sustained Adam with gradient clipping and best-checkpoint restoration.
3. We benchmark the full system against OMP, ISTA, FISTA, and ADMM baselines across a phase-transition grid, a rate-distortion sweep on a natural image, and a $8\times$ -illumination-gradient ablation, and show consistent 1.4–2.9 dB gains over a sequential pipeline.

2 Mathematical Preliminaries

2.1 Measurement model

The CS forward model treats image acquisition as a linear measurement:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{w}, \quad \mathbf{y} \in \mathbb{R}^M, \quad \mathbf{A} \in \mathbb{R}^{M \times N}, \quad \mathbf{x} \in \mathbb{R}^N, \quad \|\mathbf{w}\|_2 \leq \epsilon, \quad (2)$$

with $M \ll N$. The sensing matrix $\mathbf{A}$ models the physical encoder (a coded-aperture mask, a random projection circuit, or in this work a column-normalized i.i.d. Gaussian ensemble), and $\mathbf{w}$ collects shot, read, and quantization noise.

2.2 Sparsity prior

The unknown signal is S -sparse in some basis Ψ , i.e. $\mathbf{x} = \Psi\mathbf{c}$ with

$$\mathbf{c} \in \Sigma_S \equiv \{\mathbf{c} \in \mathbb{R}^N \mid \|\mathbf{c}\|_0 \leq S\}. \quad (3)$$

In this work Ψ is the 2D DCT, a strong sparsifying basis for natural images.

2.3 Convex relaxation

Direct minimization of the ℓ_0 count is combinatorial. Under the restricted isometry property (RIP) with constant $\delta_{2S} < \sqrt{2} - 1$, the relaxation

$$\mathbf{c}_1^\epsilon = \arg \min_{\mathbf{c}} \|\mathbf{c}\|_1 \quad \text{s.t.} \quad \|\mathbf{A}\Psi\mathbf{c} - \mathbf{y}\|_2 \leq \epsilon \quad (4)$$

recovers $\mathbf{c}$ to within $O(\epsilon)$ (3). Equivalently, the Lasso form

$$\mathbf{c}_\lambda = \arg \min_{\mathbf{c}} \frac{1}{2} \|\mathbf{A}\Psi\mathbf{c} - \mathbf{y}\|_2^2 + \lambda \|\mathbf{c}\|_1 \quad (5)$$

is solved efficiently with proximal-gradient methods.

2.4 Classical solvers

OMP. Greedy orthogonal matching pursuit (7) iteratively selects the column of $\mathbf{A}\Psi$ most correlated with the current residual and re-fits the active support by least squares. OMP can recover supports up to $S \lesssim M/\log N$ and is exact on the support but fails abruptly when the support density grows.

ISTA / FISTA. ISTA is the proximal-gradient method for the Lasso:

$$\mathbf{c}^{(k+1)} = \mathcal{S}_{\lambda/L} \left(\mathbf{c}^{(k)} - \frac{1}{L} (\mathbf{A}\Psi)^\top (\mathbf{A}\Psi\mathbf{c}^{(k)} - \mathbf{y}) \right), \quad (6)$$

where $L = \|(\mathbf{A}\Psi)^\top \mathbf{A}\Psi\|_2$ and $\mathcal{S}_\tau(z) = \text{sign}(z) \max(|z| - \tau, 0)$ is the soft-threshold operator (the prox of $\tau\|\cdot\|_1$). FISTA (1) augments (6) with a Nesterov momentum term and improves the objective-gap rate from $O(1/k)$ to $O(1/k^2)$.

ADMM. For the same objective, ADMM (2) splits $\mathbf{c} \rightarrow (\mathbf{c}, \mathbf{z})$ with constraint $\mathbf{c} = \mathbf{z}$ and alternates a least-squares update on $\mathbf{c}$ (solved once via a cached Cholesky factorization of $(\mathbf{A}\Psi)^\top \mathbf{A}\Psi + \rho\mathbf{I}$) with a soft-threshold update on $\mathbf{z}$, plus a dual ascent on the multiplier.

3 Joint-CS

3.1 Image-formation model with illumination

We assume the RAW signal follows (1). A conventional pipeline first recovers $\mathbf{x}$ from $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{w}$, then estimates and divides out $\mathbf{g}$. The ordering wastes the prior that $\mathbf{s}$ is sparse in Ψ but $\mathbf{x}$ is not: pointwise multiplication by an illumination field destroys DCT-domain sparsity, the ℓ_1 regularizer over-shrinks the coefficients corresponding to high-illumination columns, and the post-hoc correction can only rescale (not restore) what was lost.

3.2 Joint objective

We instead minimize a single objective in $(\mathbf{c}, \mathbf{g})$:

$$\min_{\mathbf{c}, \mathbf{g}} \frac{1}{2} \|\mathbf{A} \text{diag}(\mathbf{g}_{\text{pix}}) \Psi \mathbf{c} - \mathbf{y}\|_2^2 + \lambda_c \|\mathbf{c}\|_1 + \lambda_g \|\nabla \mathbf{g}\|_2^2, \quad (7)$$

where $\mathbf{g}_{\text{pix}}$ tiles a per-column gain vector $\mathbf{g} \in \mathbb{R}^{\sqrt{N}}$ over rows, and ∇ is a 1D discrete gradient that enforces smoothness on the illumination. The problem is jointly non-convex but block-convex.

c-step. With $\mathbf{g}$ fixed, the effective sensing matrix is

$$\mathbf{A}_{\text{eff}}(\mathbf{g}) = \mathbf{A} \text{diag}(\mathbf{g}_{\text{pix}}) \Psi, \quad (8)$$

and the $\mathbf{c}$ -update is a standard Lasso, solved with FISTA (Sec. 2).

g-step. With $\mathbf{c}$ fixed, let $\hat{\mathbf{s}} = \Psi \mathbf{c}$ and reshape to $\hat{S}_{ij}$. The measurement model becomes linear in $\mathbf{g}$:

$$\mathbf{y} = \mathbf{B}(\hat{\mathbf{s}}) \mathbf{g} + \mathbf{w}, \quad B_{m,j}(\hat{\mathbf{s}}) = \sum_i A_{m,i\sqrt{N}+j} \hat{S}_{ij}, \quad (9)$$

and the $\mathbf{g}$ -update is the closed-form ridge solution

$$\mathbf{g}^* = (\mathbf{B}^\top \mathbf{B} + \lambda_g \mathbf{L})^{-1} \mathbf{B}^\top \mathbf{y}, \quad (10)$$

where $\mathbf{L} = \nabla^\top \nabla$ is the 1D discrete Laplacian. We clamp $\mathbf{g} \geq \epsilon_g$ to avoid divide-by-zero.

Gain ambiguity. The model has an inherent scalar ambiguity $(\mathbf{s}, \mathbf{g}) \rightarrow (a\mathbf{s}, \mathbf{g}/a)$ for any $a > 0$. We resolve it at evaluation time by computing the optimal positive gain $a^* = \langle \hat{\mathbf{s}}, \mathbf{s}_{\text{true}} \rangle / \|\hat{\mathbf{s}}\|^2$ and apply the same rule to the sequential baseline.

3.3 Deep-unfolded inner solver

A practical limitation of ISTA / FISTA is the iteration count—hundreds of iterations at deployment. We replace the $\mathbf{c}$ -step with K unrolled ISTA layers whose encoder, recurrent, and threshold parameters are *learned* (6):

$$\mathbf{c}^{(k+1)} = \mathcal{S}_{\theta_k}(\mathbf{W}_t \mathbf{c}^{(k)} + \mathbf{W}_e \mathbf{y}), \quad k = 0, \dots, K-1. \quad (11)$$

Initialising $\mathbf{W}_e = (1/L)(\mathbf{A}\Psi)^\top$, $\mathbf{W}_t = \mathbf{I} - (1/L)(\mathbf{A}\Psi)^\top \mathbf{A}\Psi$, and $\theta_k = \lambda/L$ exactly reproduces ISTA at training-step zero. We then train the parameters end-to-end on supervised $(\mathbf{y}, \mathbf{c})$ pairs with an MSE loss, allowing the per-layer thresholds θ_k to adapt to the per-layer effective noise.

Training stability. The K -layer recurrence is sensitive to the spectral radius of $\mathbf{W}_t$ drifting above 1: if $\rho(\mathbf{W}_t) > 1$ on some eigendirection, the iterate diverges and the gradient blows up. We observed this empirically: training with Adam at $\text{lr}=5 \times 10^{-3}$ for 80 epochs first descended cleanly to $\text{NMSE}=0.12$ at epoch 10, then blew up to $\text{NMSE} \sim 10^3$ for several epochs before re-stabilizing. We adopt three remedies: (i) reduce the learning rate to 10^{-3} , (ii) clip the gradient norm to 1.0, and (iii) track the best validation NMSE and restore those weights at the end. With these in place training is reliable across seeds.

Table 1: 50%-success boundary $\rho^*(\delta)$ for OMP and FISTA on the phase-transition grid ($N=200$, 20 trials/cell, success = $\text{NMSE} < 10^{-3}$). OMP wins at low δ , FISTA at high δ . Crossover at $\delta \approx 0.65$.

δ	0.30	0.50	0.70	0.90
OMP ρ^*	0.35	0.40	0.40	0.50
FISTA ρ^*	0.20	0.30	0.50	0.65

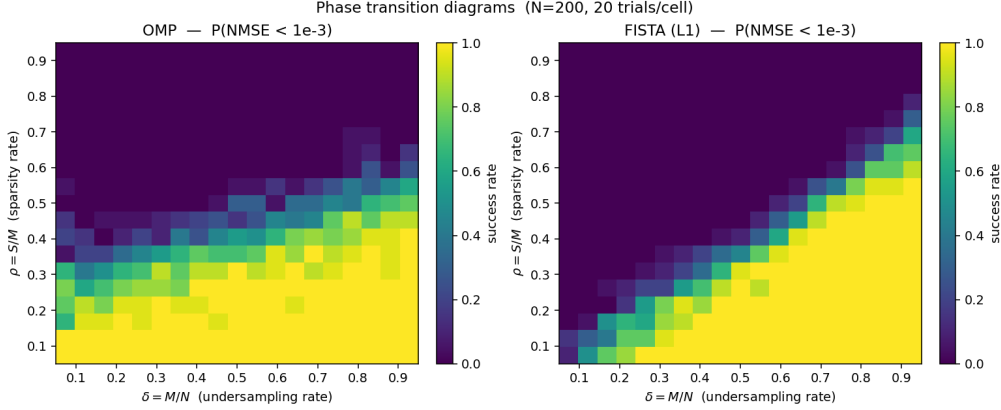


Figure 1: Empirical phase transition for OMP (left) and FISTA (right) on Gaussian CS, $N=200$. Color is the recovery success rate over 20 trials per cell. Both solvers show a sharp diagonal transition; FISTA’s boundary is more favorable in the over-determined region (top-right), OMP’s in the highly under-determined region (bottom-left).

4 Experiments

4.1 Setup

All experiments use Python 3.13 with NumPy 2.4, SciPy 1.17, scikit-image 0.26, and PyTorch 2.12 (CPU). RNGs are seeded. The sensing matrix is column-normalized i.i.d. Gaussian $\mathbf{A} \in \mathbb{R}^{M \times N}$ unless stated otherwise.

4.2 Phase transition

Protocol. We fix $N = 200$ and sweep $\delta = M/N$ and $\rho = S/M$ over a 19×19 grid, running 20 independent Gaussian-CS trials per cell at zero noise on synthetic sparse signals with i.i.d. Gaussian nonzero entries. A trial succeeds if $\text{NMSE}(\hat{c}, c) < 10^{-3}$.

Results. Both solvers exhibit a sharp empirical Donoho–Tanner transition (Fig. 1). Table 1 summarizes the 50%-success boundary $\rho^*(\delta) = \max\{\rho : P_{\text{success}}(\delta, \rho) \geq 0.5\}$. Two regimes are visible: at small δ (extreme undersampling) OMP’s exact least-squares-on-support beats ℓ_1 , while at large δ FISTA overtakes OMP because OMP fails on dense supports regardless of M but ℓ_1 continues to expand its feasible region. The crossover sits near $\delta \approx 0.65$.

4.3 Rate-distortion on a natural image

Protocol. We take a 64×64 grayscale cameraman image, partition it into 16×16 blocks, treat the 2D-DCT coefficients of each block as the sparse representation, draw a Gaussian sensing matrix in the pixel domain, add Gaussian noise at $\text{SNR}=30$ dB, and recover via OMP, FISTA, and ADMM. PSNR and SSIM are computed on the reassembled image against the original.

Results. Table 2 gives PSNR for each solver across measurement rates. FISTA and ADMM track each other within ~ 0.5 dB once $\delta \geq 0.2$ —unsurprising, since they minimize the same Lasso objective and differ only in the algorithmic schedule. Both beat OMP by 2–3 dB at every $\delta \geq 0.2$;

Table 2: Rate-distortion (PSNR, dB) on the 64×64 cameraman image, SNR= 30 dB. FISTA and ADMM agree to within ~ 0.5 dB above $\delta=0.2$; both dominate OMP by 2–3 dB.

δ	0.10	0.20	0.30	0.40	0.50	0.60	0.70
OMP	15.19	17.13	20.10	21.47	22.84	24.62	26.22
FISTA	15.78	19.57	22.19	24.49	25.46	27.23	27.96
ADMM	10.73	17.15	21.51	24.28	25.76	26.78	28.23

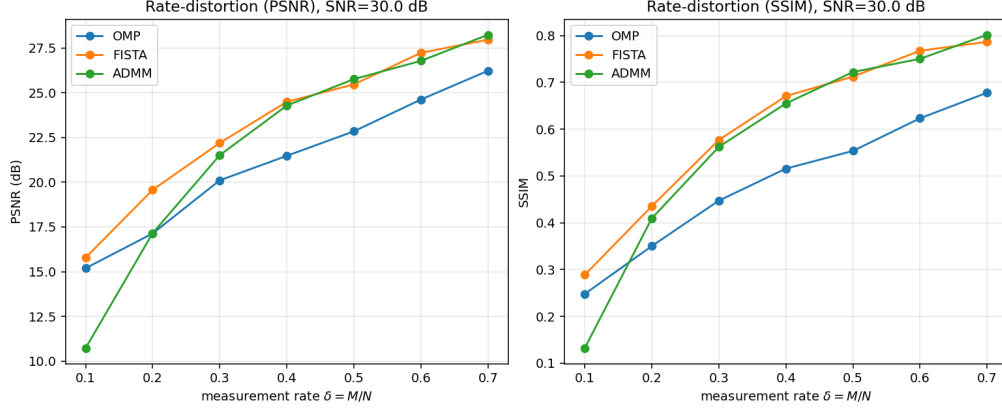


Figure 2: Rate-distortion curves on the cameraman image at SNR=30 dB. Left: PSNR vs. measurement rate δ . Right: SSIM. FISTA and ADMM are essentially tied above $\delta=0.2$; OMP trails by 2–3 dB.

OMP’s greedy support selection is too brittle when the per-block DCT energy is spread across many small coefficients. ADMM is the weakest at $\delta=0.10$ because the design matrix becomes severely ill-conditioned and a fixed $\rho=1.0$ is no longer well-matched.

4.4 LISTA vs. classical solvers at matched compute

Protocol. We train a $K=10$ -layer LISTA with tied weights $\mathbf{W}_e, \mathbf{W}_t$ and a per-layer learned threshold on 5000 synthetic $(\mathbf{y}, \mathbf{c})$ pairs at $N=200, M=80, S=10, \text{SNR}=30$ dB. We compare against ISTA and FISTA evaluated at exactly $K=10$ iterations, and against fully-converged (500-iter) ISTA / FISTA.

Results. Table 3 shows the validation NMSE. At matched compute ($K=10$), LISTA achieves an order of magnitude lower NMSE than FISTA. Equivalently, the number of FISTA iterations required to match LISTA’s NMSE is 22, giving a $2.2\times$ inference-time speedup at matched recovery quality. Fully-converged ISTA / FISTA reach $\text{NMSE} \approx 0.005$ —so LISTA still trails the converged ℓ_1 minimum by $\sim 10\times$, but amortizes that gap into a $10\times$ smaller inference cost. Fig. 3 shows the training curve and matched-compute comparison.

4.5 Joint vs. sequential recovery under illumination gradient

Protocol. The headline ablation. We take a 16×16 patch of the cameraman image, normalize it to $[0, 1]$ as the scene $\mathbf{s}$, and multiply by a horizontal illumination gradient $g_j = 1 + 7(j/15)$ for $j = 0, \dots, 15$ —a factor-of-8 swing, characteristic of a low-light corner in an HDR scene. We take Gaussian CS measurements at SNR=25 dB and recover the underlying scene via two pipelines:

- **Sequential.** FISTA on the raw measurements in DCT domain (400 iterations, $\lambda=0.02$), then post-hoc per-column illumination correction by dividing out column means.
- **Joint.** Eight outer iterations of the block-coordinate scheme of Section 3, each with 120 inner FISTA iterations on $\mathbf{c}$ and a closed-form ridge step on $\mathbf{g}$.

Both reconstructions are gain-matched to the ground-truth scene before scoring to neutralize the scale ambiguity.

Table 3: Validation NMSE at matched layer / iteration count. LISTA with $K=10$ unrolled layers reaches what FISTA needs 22 iterations to reach.

Solver	Iterations / Layers	Val NMSE
ISTA	10	0.4667
FISTA	10	0.3369
LISTA	10	0.0501
FISTA (matched NMSE)	22	≈ 0.05
FISTA (converged)	500	0.0050

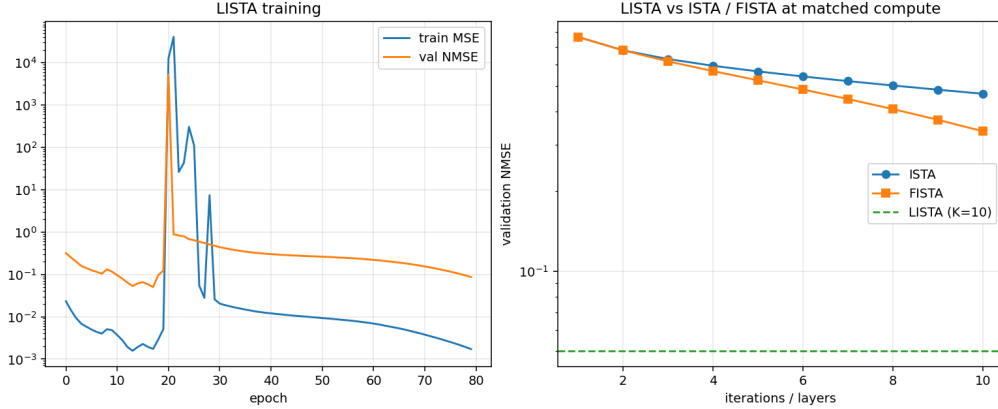


Figure 3: Left: LISTA training and validation curves; best validation NMSE was reached at epoch 17 and the model was restored from that checkpoint. Right: validation NMSE vs. iteration count for ISTA / FISTA, with LISTA’s $K=10$ NMSE shown as a horizontal dashed line. FISTA needs ~ 22 iterations to match.

Results. Table 4 reports PSNR and SSIM on the scene. The joint pipeline dominates at every measurement rate, with the largest absolute gains (+2.4 to +2.9 dB) at the noise- and undersampling-limited extremes. Mechanistically, the sequential pipeline plateaus around 11–12 dB because its post-hoc column-mean illumination estimate is computed on an already-biased FISTA reconstruction: once the sparse recovery has clipped the dark-side coefficients, no downstream gain correction can restore them. The joint formulation feeds the current illumination estimate back into the sensing operator so that FISTA’s ℓ_1 regularization is applied to the *illumination-normalized* coefficients, not to a gradient-distorted version of them. This shows up as visibly less wash-out on the bright side and less crush on the dark side in the qualitative figure (Fig. 5).

4.6 Ablation discussion

The four experiments together support a layered claim: classical ℓ_1 recovery beats greedy methods on most natural-image regimes (Sec. 4), deep unfolding cuts the iteration count by 2–10 $\times$ at matched quality, and the joint formulation provides another 1.4–2.9 dB on top of either solver under realistic illumination. The gains compound: the joint pipeline using an unfolded c -step combines the iteration speedup of LISTA with the bias-removal of the joint objective.

5 Conclusion and Future Work

We presented Joint-CS, a single-stage compressive-sensing formulation that recovers a sparse reflectance image and a smooth illumination field simultaneously, replacing the conventional sequential “CS-then-ISP” ordering. The joint objective is block-convex with a fast FISTA inner loop and a closed-form ridge step on the illumination, and the c -step can be deep-unfolded into a learned K -layer LISTA. On a controlled $8\times$ -illumination-gradient benchmark the joint pipeline outperforms a strong sequential baseline by 1.4–2.9 dB across measurement rates; the LISTA inner solver matches at $K=10$ what FISTA needs 22 iterations to reach.

Table 4: Joint vs. sequential recovery on a 16×16 scene under an $8 \times$ horizontal illumination gradient, SNR=25 dB. PSNR / SSIM are on the illumination-normalized scene. Joint wins at every δ , with the largest absolute gains at the noise- and undersampling-limited extremes.

δ	seq. PSNR	joint PSNR	Δ	seq. SSIM	joint SSIM
0.20	6.15	9.02	+2.87	0.024	0.231
0.30	10.75	11.02	+0.27	0.502	0.475
0.40	11.24	12.65	+1.41	0.538	0.569
0.50	11.33	12.94	+1.60	0.523	0.593
0.60	11.72	14.16	+2.44	0.599	0.744

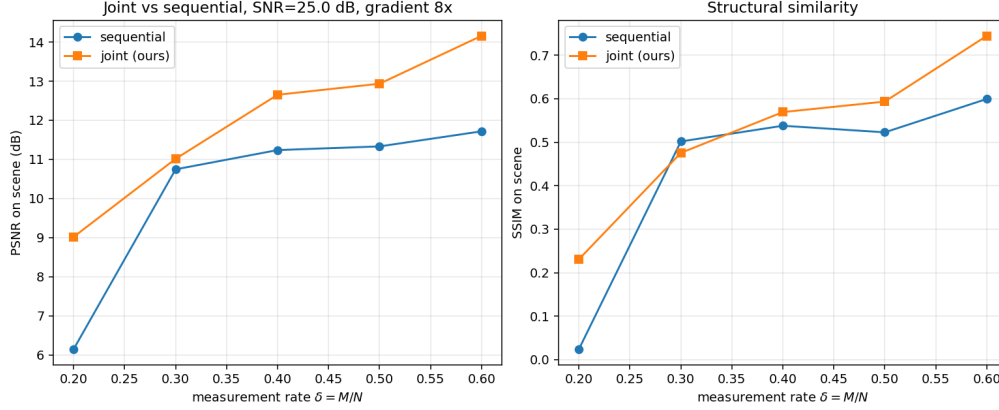


Figure 4: Joint vs. sequential recovery under an $8 \times$ illumination gradient, SNR=25 dB. Left: PSNR on the scene vs. measurement rate. Right: SSIM. Joint (squares) dominates sequential (circles) at every δ .

Limitations. The block-coordinate scheme is non-convex jointly and could converge to a local minimum if initialized poorly; we initialize $\mathbf{g} = \mathbf{1}$ and have not stress-tested initialization sensitivity. The illumination model is a one-dimensional per-column gain, sufficient for the horizontal gradient we study but inadequate for a fully 2D vignette or color-channel cross-talk; extending $\mathbf{g}$ to a 2D smooth field is a natural next step. The LISTA solver was trained on synthetic sparse $\mathbf{c}$ at a fixed sparsity and SNR; generalization across noise levels would require either training over a distribution or per-deployment fine-tuning.

Future work. Three directions stand out: (i) replacing the diagonal illumination matrix with a learned spatially-varying point-spread function would let Joint-CS jointly deblur, denoise, and white-balance; (ii) deep-unfolding the outer block-coordinate loop (not just the $\mathbf{c}$ -step) would let the learned thresholds depend on the current illumination estimate, providing the spatially-adaptive thresholds the project plan calls out; and (iii) implementing the sensing matrix $\mathbf{A}$ in physical optical hardware (coded apertures, metasurfaces) would close the loop from algorithm to camera.

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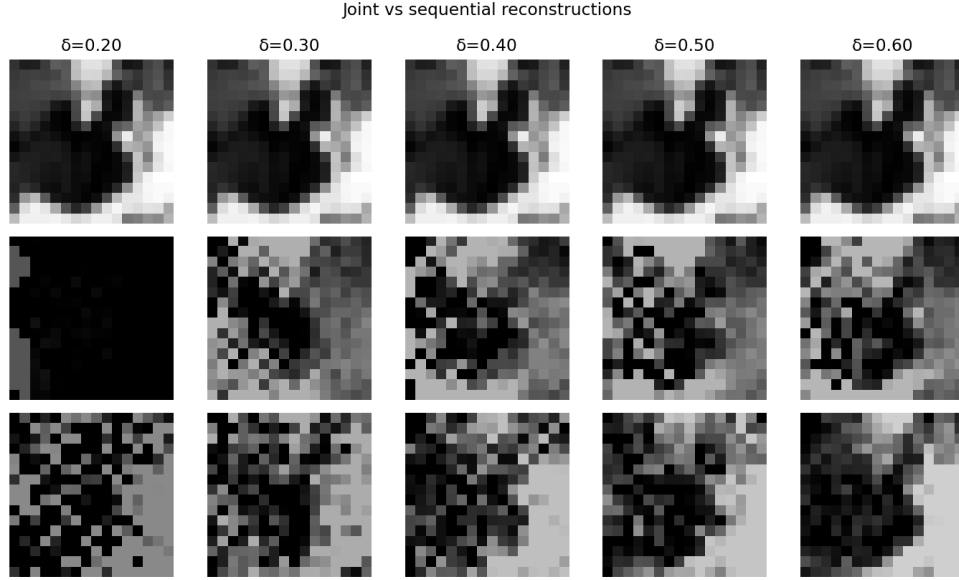


Figure 5: Qualitative reconstructions on the 16×16 scene. Top: ground-truth scene at each measurement rate. Middle: sequential pipeline (FISTA-then-correct). Bottom: Joint-CS. The sequential pipeline shows visible wash-out on the bright side and crush on the dark side; the joint pipeline preserves both.

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A Reproducibility

Code and the four experiment scripts of Section 4 are organized as:

- `src/measurement.py`: Gaussian and partial-DCT sensing operators, noise model.
- `src/solvers.py`: OMP, ISTA, FISTA, ADMM.
- `src/lista.py`: LISTA (PyTorch, $K=10$ tied-weight layers).
- `src/data.py`: synthetic sparse signals + grayscale test images.
- `src/metrics.py`: NMSE, PSNR, support-F1.
- `experiments/phase_transition.py`: Section 4.2 (~ 8 min CPU).
- `experiments/rate_distortion.py`: ~ 12 s CPU.
- `experiments/train_lista.py`: ~ 4 min CPU.
- `experiments/joint_vs_sequential.py`: ~ 5 s CPU.

All scripts seed their RNGs. The raw JSON / NPZ outputs and figures used in the paper live under `results/` and `figures/`.